

DATE:

EXPT NO:03

EDA- Data Cleaning and Data Transformation

AIM:

To inspect and analyze the Earthquake dataset using Exploratory Data Analysis (EDA) to understand the data structure, identify patterns, detect missing values, and discover relationships between different earthquake attributes.

PROCEDURE:

1. Import Pandas, NumPy, Matplotlib, and Seaborn libraries.
2. Load the Earthquake dataset using Pandas.
3. Display the first and last records using head() and tail().
4. Check the number of rows and columns using shape.
5. Display column names and data types using info().
6. Check missing values using isnull().sum().
7. Generate statistical summaries using describe().
8. Check and identify duplicate records.
9. Analyze important earthquake-related attributes.
10. Create suitable graphs such as histograms, bar charts, and scatter plots.
11. Generate a correlation heatmap to identify relationships between numerical variables.
12. Interpret the findings and identify important patterns in the Earthquake dataset.

CODE:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

file_path = r"C:\Users\babyn\Downloads\archive\earthquake_data.xlsx"
df = pd.read_excel(file_path)
print("Dataset Preview:")
print(df.head())

# 1. Identifying Missing Values
print("\nMissing Values in Each Column:")
print(df.isnull())

# 2. Count Missing Values
print("\nTotal Missing Values:")
print(df.isnull().sum())
```

```
# Import required libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import MinMaxScaler

# Load Dataset
file_path = r"C:\Users\babyn\Downloads\archive\earthquake_data.xlsx"
df = pd.read_excel(file_path)
print("Original Dataset:")
print(df.head())
print("\nDataset Information:")
print(df.info())

# 1. Identify Missing Values
print("\nMissing Values in Dataset:")
print(df.isnull())

# 2. Count Missing Values
print("\nTotal Missing Values in Each Column:")
print(df.isnull().sum())

# 3. Replace Missing Values
# Numerical columns - replace missing values with mean
numeric_columns = df.select_dtypes(include=np.number).columns
for col in numeric_columns:
    df[col] = df[col].fillna(df[col].mean())

# Categorical columns - replace missing values with mode
categorical_columns = df.select_dtypes(include="object").columns
for col in categorical_columns:
    df[col] = df[col].fillna(df[col].mode()[0])

print("\nMissing Values After Replacement:")
print(df.isnull().sum())

# 4. Detect Duplicate Records
print("\nNumber of Duplicate Rows:")
print(df.duplicated().sum())

# Remove duplicate records
df = df.drop_duplicates()
print("\nDuplicate Rows After Removing:")
print(df.duplicated().sum())

# 5. Detect Outliers using IQR Method
print("\nOutlier Detection:")
outlier_result = {}
for col in numeric_columns:
    Q1 = df[col].quantile(0.25)
```

```

Q3 = df[col].quantile(0.75)
IQR = Q3 - Q1
lower_bound = Q1 - (1.5 * IQR)
upper_bound = Q3 + (1.5 * IQR)
outliers = df[(df[col] < lower_bound) | (df[col] > upper_bound)]
outlier_result[col] = len(outliers)
print(outlier_result)

# 6. Outlier Visualization
for col in numeric_columns:
    plt.figure(figsize=(6, 4))
    sns.boxplot(x=df[col])
    plt.title("Outlier Detection: " + col)
    plt.show()

# 7. Data Cleaning
# Remove spaces from column names
df.columns = df.columns.str.strip()
# Convert column names into lowercase
df.columns = df.columns.str.lower()
print("\nCleaned Column Names:")
print(df.columns)

# 8. Data Transformation
# Normalization using MinMaxScaler
scaler = MinMaxScaler()
df[numeric_columns] = scaler.fit_transform(df[numeric_columns])
print("\nTransformed Dataset:")
print(df.head())

# 9. Save Cleaned Dataset
output_file = "cleaned_earthquake_data.xlsx"
df.to_excel(output_file, index=False)
print("\nCleaned Dataset Saved Successfully!")
print("File Name:", output_file)

```

OUTPUT:

Original Dataset:

	mmi	alert	tsunami	sig	net	nst	dmin	gap	magType	depth	latitude	longitude	location	continent	country	title	magnitude	date_time	cdi
0	M 7.0	-	18 km SW of Malango, Solomon Islands							7.0	22-11-2022 02:03								8
7	green		1	768	us	117	0.509	17.0	mww	14.000	-9.7963								159.596
													Malango, Solomon Islands	Oceania	Solomon Islands				
1	M 6.9	-	204 km SW of Bengkulu, Indonesia							6.9	18-11-2022 13:37								4
4	green		0	735	us	99	2.229	34.0	mww	25.000	-4.9559								100.738
													Bengkulu, Indonesia		NaN				
2																			
																M 7.0 -			
3	green		1	755	us	147	3.125	18.0	mww	579.000	-20.0508								3
													NaN	Oceania		Fiji			
3																			
													M 7.3	-	205 km ESE of Neiafu, Tonga				5

5	green	1	833	us	149	1.865	21.0	mww	37.000	-19.2918	-172.129
Neiafu, Tonga		NaN				NaN					
4							M 6.6 -		6.6	09-11-2022 10:14	0
2	green	1	670	us	131	4.998	27.0	mww	624.464	-25.5948	178.278
NaN		NaN		NaN							

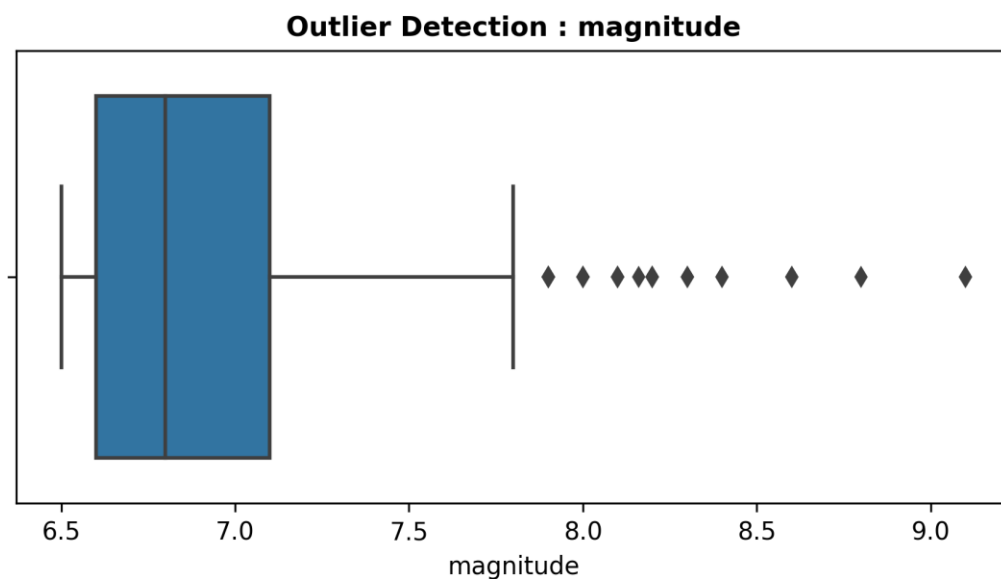
Missing Values in Dataset:

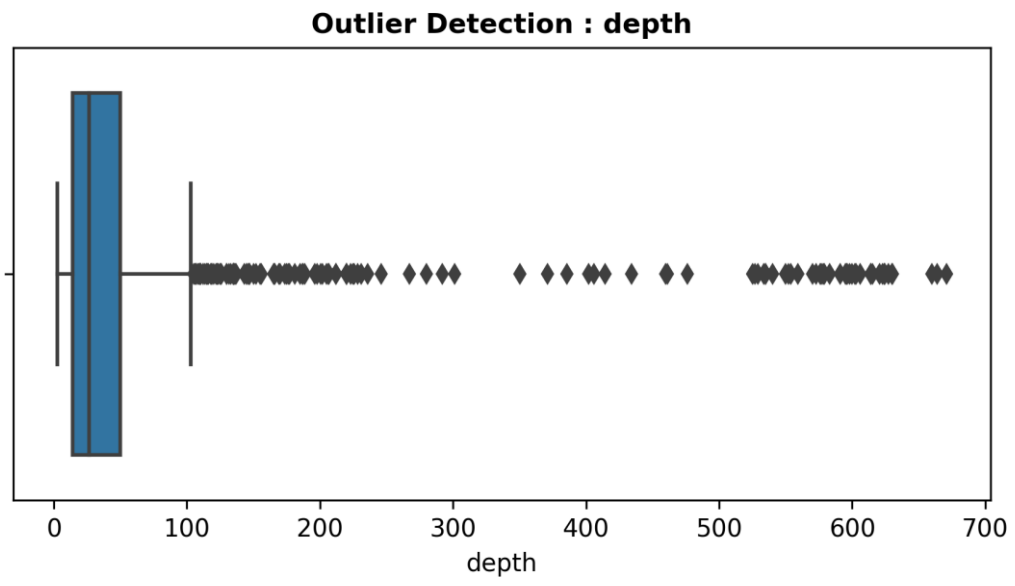
Total Missing Values in Each Column:

title	0
magnitude	0
date_time	0
cdi	0
mmi	0
alert	367
tsunami	0
sig	0
net	0
nst	0
dmin	0
gap	0
magType	0
depth	0
latitude	0
longitude	0
location	5
continent	576
country	298
dtype: int64	

Outlier Detection:

```
{'magnitude': 38, 'cdi': 0, 'mmi': 0, 'tsunami': 0, 'sig': 42, 'nst': 34, 'dmin': 48, 'gap': 25, 'depth': 86, 'latitude': 0, 'longitude': 0}
```





RESULT: